



6S battery problems

Apple announces a repair program for some iPhone 6S models regarding battery issues. <http://dgit.in/6Sapleissue>



HYUNDAI

Hyundai electric buses

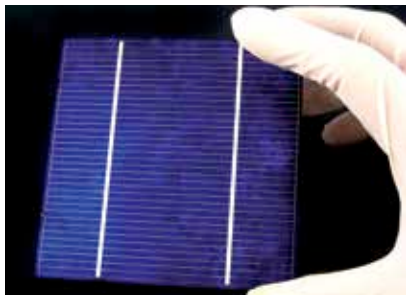
South Korean automaker, Hyundai is preparing on mass-producing electric buses by 2017 <http://dgit.in/hyndiEbus>

Tomorrow's Tech

But there is some good news: newer materials make use of smaller and cheaper crystals like copper-indium-gallium-selenide. These can even be shaped into thin flexible films. This is called the “thin-film” solar technology.

But the catch is that this technology isn't as good when it comes to turning light into electricity compared to silicon. Even in the U.S – the wealthiest nation in the world, solar energy accounts for just a small portion of their complete electricity generation – the reason being, of course, the expense involved. Generating electricity from coal is cheaper but highly polluting.

To get a perspective on how expensive solar power is at the moment, it would be about five times as expensive



Efficient but costly

as what you pay for the electricity which comes out of the power outlets in your home.

This cost-factor has been a major reason why the world hasn't been able to adopt solar panels widely even though they have been considered as a way to replace fossil fuels.

Is the action of solar cells similar to photosynthesis?

One comparison that many laymen make is between the functioning of solar cells and photosynthesis. But that's not entirely an accurate comparison. For instead of just creating electricity, plants absorb light and use the electron to alter a chemical bond within the plant-meaning they generate fuel for their consumption.

Now, it would be super-cool if scientists were able to create a solar cell which, instead of creating electricity, made fuel. Aside from the fuel-advan-



Photosynthesis: a different ballgame

tage, it would also help deal with the effects of global warming because by mimicking a plant, the solar cell would also be taking in the CO₂ from the air to generate the fuel.

The power efficiency of a cell

Since we use solar cells to capture sunlight and turn it into electricity, it's worth looking at the power efficiency of a typical crystalline silicon cell.

The power efficacy of such a cell is in the range of 22 to 23 percent (meaning, 22 or 23 percent of light that strikes a cell is converted into electricity). But that's the case with the industrial variant. The ones which are affordable for an individual to put on the rooftop have a power efficacy that lies somewhere within the 15 to 18 percent range. The most efficient variants are the ones that are used in satellites – these could have power efficiencies which approach a whopping 50 percent.



More power to you

Powering an entire island with solar energy

Solar City Corporation is one of the more significant players in the solar energy products arena. Based in America, the company recently merged with the Elon Musk-led Tesla. Even though the merger was completed

mere weeks back, a glimpse of what such a union could bring to the world has already been shown – in the way of an entire island that's powered with solar energy.

The island in question is the small island of Ta'u in American Samoa. It's over 4,000 miles from the West Coast of the United States. The island is a 17-square mile volcanic island that has just a single road that runs through it.

The project made use of the prowess of SolarCity power generation and Tesla's Powerpack-based storage. Before this, the island was relying a lot on a diesel generator grid.

The reliance on diesel as a fuel to generate power isn't all that unique. Many island communities rely on burning fossil fuel to maintain an electrical grid. However, using a fuel which has to be shipped in at great costs from long distances poses a large number of problems. Also, such a scenario causes a lot of temporary blackouts. The costs of solar generally can be recouped only in a long time once the grid power becomes convenient and easily available. However, the financial benefits get accrued way more quickly when you are shipping in fuel and burning it as the sole source of energy.

For the Ta'u grid project, a SolarCity panel array was created that's made of more than 5,300 panels with the



An island charged by the sun

capability of generating 1.4 megawatts of solar capacity. This was clubbed with a Tesla Powerpack commercial battery storage facility that has a capability of storing 6 megawatt hours of reserve capability in 60 Powerpacks. The resultant micro-grid could bring three full days worth of power for the island's